Q1 - 2024 (04 Apr Shift 1)

In an experiment to measure focal length (f) of convex lens, the least counts of the measuring scales for the position of object (u) and for the position of image (v) are $\Delta \mathrm{u}$ and $\Delta \mathrm{v}$, respectively. The error in the measurement of the focal length of the convex lens will be:

(1) $2f\left[\frac{\Delta \mathrm{u}}{\mathrm{u}}+\frac{\Delta \mathrm{v}}{\mathrm{v}}\right]$

(2) $\frac{\Delta \mathrm{u}}{\mathrm{u}}+\frac{\Delta \mathrm{v}}{\mathrm{v}}$

(3) $f^2\left[\frac{\Delta \mathrm{u}}{\mathrm{u}^2}+\frac{\Delta \mathrm{v}}{\mathrm{v}^2}\right]$

(4) $f\left[\frac{\Delta \mathrm{u}}{\mathrm{u}}+\frac{\Delta \mathrm{v}}{\mathrm{v}}\right]$

Q2 - 2024 (04 Apr Shift 1)

Two forces $\bar{F}_1$ and $\bar{F}_2$ are acting on a body. One force has magnitude thrice that of the other force and the resultant of the two forces is equal to the force of larger magnitude. The angle between $\vec{F}_1$ and $\vec{F}_2$ is $\cos^{-1}\left(\frac{1}{n}\right)$. The value of $|n|$ is _____.

Q3 - 2024 (05 Apr Shift 1)

The angle between vector $\vec{Q}$ and the resultant of $(2\vec{Q}+2\vec{P})$ and $(2\vec{Q}-2\vec{P})$ is :

(1) $\tan^{-1}\frac{(2\vec{Q}-2\vec{P})}{2\vec{Q}+2\vec{P}}$

(2) 0°

(3) $\tan^{-1}(\mathrm{P}/\mathrm{Q})$

(4) $\tan^{-1}(2\mathrm{Q}/\mathrm{P})$

Q4 - 2024 (05 Apr Shift 1)

Time periods of oscillation of the same simple pendulum measured using four different measuring clocks were recorded as 4.62 s, 4.632 s, 4.6 s and 4.64 s. The arithmetic mean of these readings in correct significant figure is :

(1) 5 s

(2) 4.623 s

(3) 4.6 s

(4) 4.62 s

Q5 - 2024 (05 Apr Shift 2)

A particle moves in $x-y$ plane under the influence of a force $\vec{F}$ such that its linear momentum is $\overrightarrow{\mathrm{p}}(\mathrm{t})=\hat{i}\cos(\mathrm{kt})-\hat{j}\sin(\mathrm{kt})$. If k is constant, the angle between $\overrightarrow{\mathrm{F}}$ and $\overrightarrow{\mathrm{p}}$ will be :

(1) $\frac{\pi}{4}$

(2) $\frac{\pi}{6}$

(3) $\frac{\pi}{2}$

(4) $\frac{\pi}{3}$

Q6 - 2024 (06 Apr Shift 1)

To find the spring constant (k) of a spring experimentally, a student commits 2% positive error in the measurement of time and 1% negative error in measurement of mass. The percentage error in determining value of k is :

(1) 5%

(2) 1%

(3) 3%

(4) 4%

Q7 - 2024 (06 Apr Shift 1)

For three vectors $\vec{A}=(-x\hat{i}-6\hat{j}-2\hat{k})$, $\vec{B}=(-\hat{i}+4\hat{j}+3\hat{k})$ and $\vec{C}=(-8\hat{i}-\hat{j}+3\hat{k})$, if $\vec{A}\cdot(\vec{B}\times\vec{C})=0$, then value of x is _________

Q8 - 2024 (08 Apr Shift 1)

In an expression $a\times10^{\mathrm{b}}$;

(1) b is order of magnitude for $a\geq5$

(2) b is order of magnitude for $a \leq 5$

(3) a is order of magnitude for $b \leq 5$

(4) b is order of magnitude for $5 < a \leq 10$

Q9 - 2024 (08 Apr Shift 1)

Young's modulus is determined by the equation given by $\mathrm{Y} = 49000 \frac{\mathrm{m}}{\mathrm{l}} \frac{\mathrm{dyn}}{\mathrm{cm}^2}$ where M is the mass and l is the extension of wire used in the experiment. Now error in Young modules (Y) is estimated by taking data from $M - l$ plot in graph paper. The smallest scale divisions are 5 g and 0.02 cm along load axis and extension axis respectively. If the value of M and l are 500 g and 2 cm respectively then percentage error of Y is :

(1) 0.5%

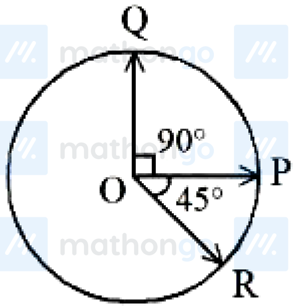
(2) 2%

(3) 0.02%

(4) 0.2%

Q10 - 2024 (08 Apr Shift 1)

Three vectors $\overrightarrow{\mathrm{OP}}, \overrightarrow{\mathrm{OQ}}$ and $\overrightarrow{\mathrm{OR}}$ each of magnitude A are acting as shown in figure. The resultant of the three vectors is $\mathrm{A}\sqrt{x}$. The value of x is _________ .



Q11 - 2024 (09 Apr Shift 1)

If $\vec{a}$ and $\vec{b}$ makes an angle $\cos^{-1}\left(\frac{5}{9}\right)$ with each other, then $|\vec{a} + \vec{b}| = \sqrt{2}|\vec{a} - \vec{b}|$ for $|\vec{a}| = n|\vec{b}|$ The integer value of n is ____

Q12 - 2024 (09 Apr Shift 2)

The resultant of two vectors $\vec{A}$ and $\vec{B}$ is perpendicular to $\vec{A}$ and its magnitude is half that of $\vec{B}$. The angle between vectors $\vec{A}$ and $\vec{B}$ is ______ $\circ$.

Answer Key

Q1 (3)	**Q2** (6)	**Q3** (2)	**Q4** (3)
Q5 (3)	**Q6** (1)	**Q7** (4)	**Q8** (2)
Q9 (2)	**Q10** (3)	**Q11** (3)	**Q12** (150)

Q1

$$f^{-1} = v^{-1} - u^{-1}$$
$$-f^{-2}df = -v^{-2}dv - u^{-2}du$$
$$\frac{df}{f^2} = \frac{dv}{v^2} + \frac{du}{u^2}$$
$$df = f^2\left[\frac{dv}{v^2} + \frac{du}{u^2}\right]$$

Q2

$$\left|\vec{F}_1\right| = F$$
$$\left|\overrightarrow{F}_R\right| = \left|\overrightarrow{F}_2\right| = 3\,F$$
$$F_R^2 = F_1^2 + F_2^2 + 2\,F_1\,F_2\cos\theta$$
$$9\,F^2 = F^2 + 9\,F^2 + 6\,F^2\cos\theta$$
$$\cos\theta = -\frac{1}{6}$$
$$\theta = \cos^{-1}\left(\frac{1}{-6}\right)$$
$$n = -6$$
$$|n| = 6$$

Q3

$$\overrightarrow{R} = (2\overrightarrow{Q} + 2\overrightarrow{P}) + (2\overrightarrow{Q} - 2\overrightarrow{P})$$
$$\overrightarrow{R} = 4\overrightarrow{Q}$$

Angle between $\vec{Q}$ and $\vec{R}$ is zero.

Q4

Sum of number by considering significant digi

$$\text{sum} = 4.6 + 4.6 + 4.6 + 4.6 = 18.4$$

$$\text{Arithmetic Mean} = \frac{\text{sum}}{4} = \frac{18.4}{4} = 4.6$$

Q5

$\overrightarrow{\mathrm{P}}=\cos(\mathrm{kt})\hat{\mathrm{i}}-\sin(\mathrm{kt})\hat{\mathrm{j}};|\overrightarrow{\mathrm{P}}|=1$

$\because \overrightarrow{\mathrm{P}}=\mathrm{m}\overrightarrow{\mathrm{v}}$

$\therefore \hat{\mathrm{P}}=\hat{\mathrm{v}}$

$\Rightarrow \hat{\mathrm{v}}=\cos(\mathrm{kt})\hat{\mathrm{i}}-\sin(\mathrm{kt})\hat{\mathrm{j}}$

$$\hat{\mathrm{a}}=\frac{-\mathrm{k}\sin(\mathrm{kt})\hat{\mathrm{i}}-\mathrm{k}\cos(\mathrm{kt})\hat{\mathrm{j}}}{\mathrm{k}}$$

$\Rightarrow \hat{\mathrm{a}}=-\sin kt\,\hat{\mathrm{i}}-\cos \mathrm{kt}\,\hat{\mathrm{j}}$

$\because \hat{\mathrm{F}}=\hat{\mathrm{a}}=-\sin \mathrm{kt}\,\hat{\mathrm{i}}-\cos \mathrm{kt}\,\hat{\mathrm{j}}$

$$\cos\theta=\frac{\hat{\mathrm{F}}\cdot\hat{\mathrm{P}}}{|\hat{\mathrm{F}}||\hat{\mathrm{P}}|}=-\frac{\sin \mathrm{kt}\cos \mathrm{t}+\sin \mathrm{kt}\cos \mathrm{t}}{1\times 1}=0$$

$$\Rightarrow \theta=\frac{\pi}{2}$$

Q6

$$\mathrm{T}=2\pi\sqrt{\frac{\mathrm{m}}{\mathrm{k}}}$$

$$\mathrm{T}^2\propto\frac{\mathrm{m}}{\mathrm{k}}$$

$$\frac{2\Delta \mathrm{T}}{\mathrm{T}}\%=\frac{\Delta \mathrm{m}}{\mathrm{m}}\%-\frac{\Delta \mathrm{k}}{\mathrm{k}}\%$$

$$\frac{\Delta \mathrm{k}}{\mathrm{k}}\%=\frac{\Delta \mathrm{m}}{\mathrm{m}}\%-\frac{2\Delta \mathrm{T}}{\mathrm{T}}\%$$

$$\frac{\Delta \mathrm{k}}{\mathrm{k}}\%=(-1)\%-2(2)\%=|-5\%|=5\%$$

Q7

$$\vec{B}\times\vec{C}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ -1 & 4 & 3\\ -8 & -1 & 3\end{vmatrix}=15\hat{i}-21\hat{j}+33\hat{k}$$

$$\vec{A}\cdot(\vec{B}\times\vec{C})=(-x\hat{i}-6\hat{j}-2\hat{k})\cdot(15\hat{i}-21\hat{j}+33\hat{k})$$

$0=-15x+126-66$

$15x=60$

$x=4$

Q8

$a\times 10^b$

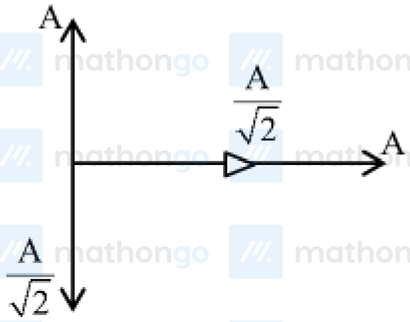
if $a\leq 5$ order is b

$a>5$ order is $b+1$

Q9

$$\frac{\Delta Y}{Y}=\frac{\Delta m}{m}+\frac{\Delta \ell}{\ell}$$
$$=\frac{5}{500}+\frac{0.02}{2}=0.01+0.01$$
$$\frac{\Delta Y}{Y}=0.02 \Rightarrow \% \frac{\Delta Y}{Y}=2\%$$

Q10

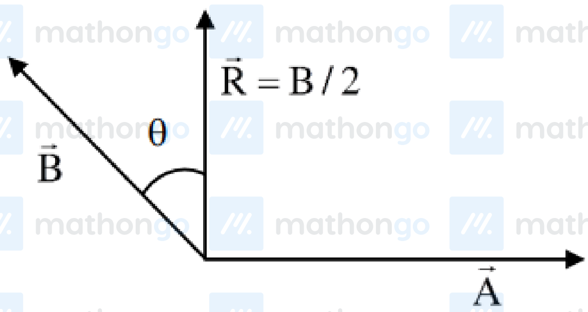


$$\vec{R}=\left(A+\frac{A}{\sqrt{2}}\right) \hat{i}+\left(A-\frac{A}{\sqrt{2}}\right) \hat{j}$$
$$|\vec{R}|=\sqrt{\left(A+\frac{A}{\sqrt{2}}\right)^{2}+\left(A-\frac{A}{\sqrt{2}}\right)^{2}}=\sqrt{3} A$$

Q11

$$\cos \theta=\frac{5}{9}$$
$$\frac{\vec{a} \cdot \vec{b}}{a b}=\frac{5}{9}$$
$$a^{2}+b^{2}+2 \vec{a} \cdot \vec{b}=2 a^{2}+2 b^{2}-4 \vec{a} \cdot \vec{b}$$
$$6 \vec{a} \cdot \vec{b}=a^{2}+b^{2}$$
$$6 \times \frac{5}{9} a b=a^{2}+b^{2}$$
$$\frac{10}{3} a b=a^{2}+b^{2} \quad \& \quad a=n b$$
$$\frac{10}{3} n b^{2}=n^{2} b^{2}+b^{2}$$
$$3 n^{2}-10 n+3=0$$
$$n=\frac{1}{3} \text{ and } n=3$$
integer value $n=3$

Q12



$\mathrm{B} \cos \theta=\frac{\mathrm{B}}{2}$

$\Rightarrow \theta=60^{\circ}$

So, angle between $\vec{A}$ & $\vec{B}$ is $90^{\circ}+60^{\circ}=150^{\circ}$